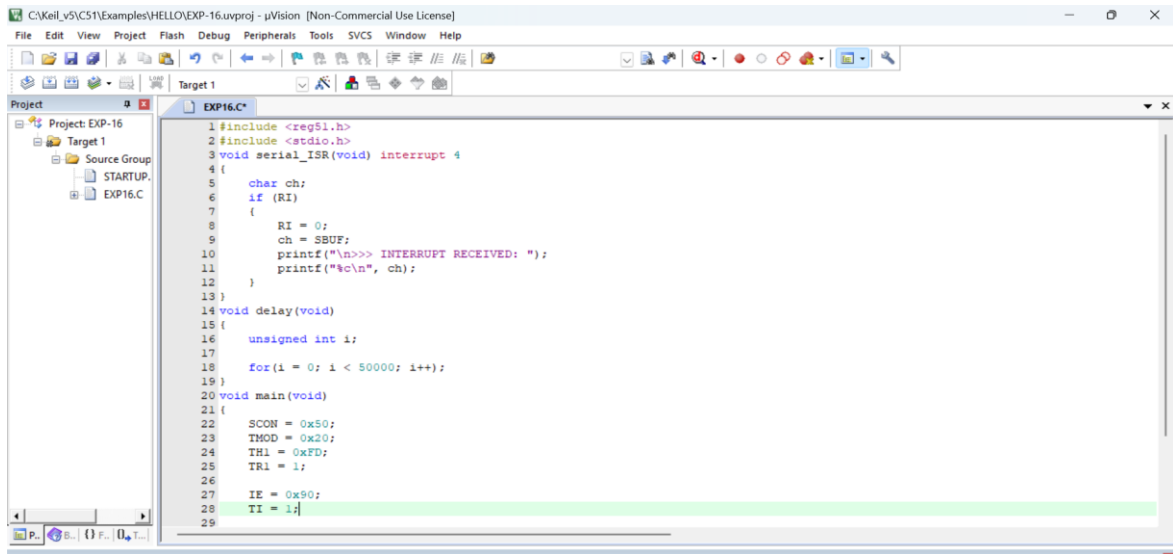
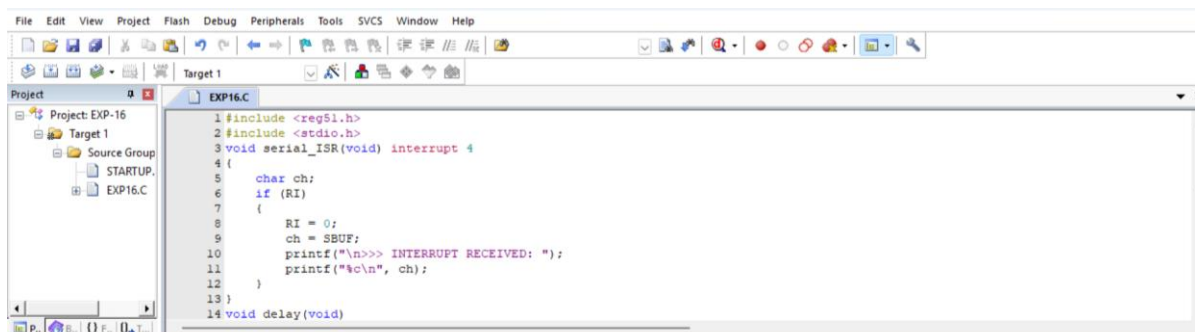


Exp No:16

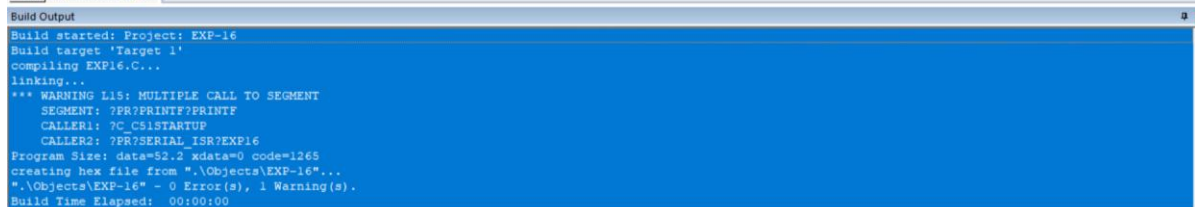
SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL



```
1#include <reg51.h>
2#include <stdio.h>
3void serial_ISR(void) interrupt 4
4{
5    char ch;
6    if (RI)
7    {
8        RI = 0;
9        ch = SBUF;
10       printf("\n>>> INTERRUPT RECEIVED: ");
11       printf("%c\n", ch);
12    }
13}
14void delay(void)
15{
16    unsigned int i;
17    for(i = 0; i < 50000; i++);
18}
19
20void main(void)
21{
22    SCON = 0x50;
23    TMOD = 0x20;
24    TH1 = 0xFD;
25    TR1 = 1;
26
27    IE = 0x90;
28    TI = 1;
29}
```



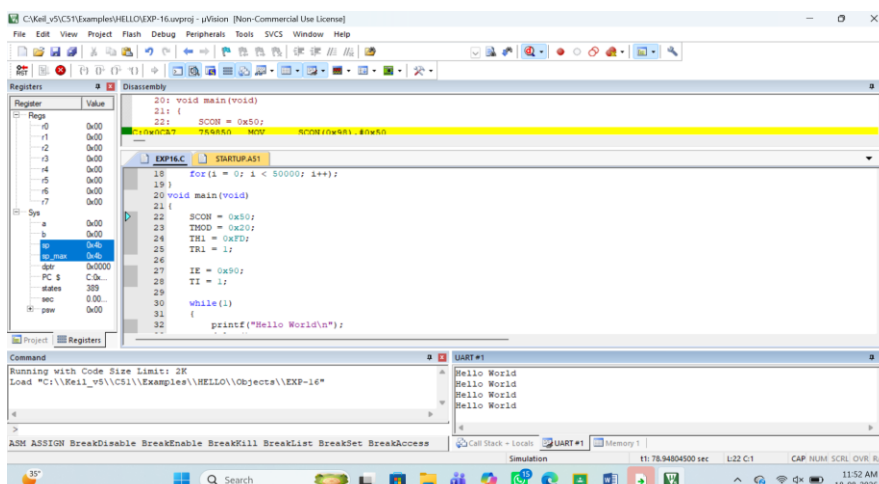
```
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3void serial_ISR(void) interrupt 4
4{
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6    if (RI)
7    {
8        RI = 0;
9        ch = SBUF;
10       printf("\n>>> INTERRUPT RECEIVED: ");
11       printf("%c\n", ch);
12    }
13}
14void delay(void)
```



```
Build started: Project: EXP-16
Build target 'Target 1'
compiling EXP16.C...
linking...
*** WARNING L15: MULTIPLE CALL TO SEGMENT
SEGMENT: ?PR?PRINTF?PRINTF
CALLER1: ?C_CS1STARTUP
CALLER2: ?PR?SERIAL_ISR?EXP16
Program Size: data=52.2 kdata=0 code=1265
creating hex file from ".\Objects\EXP-16"...
".\Objects\EXP-16" - 0 Error(s), 1 Warning(s).
Build Time Elapsed: 00:00:00
```

OUTPUT:

a) Displaying a Message:



```
20: void main(void)
21: {
22:     SCON = 0x50;
23:     TMOD = 0x20;
24:     TH1 = 0xFD;
25:     TR1 = 1;
26:
27:     IE = 0x90;
28:     TI = 1;
29:
30:     while(1)
31:     {
32:         printf("Hello World\n");
33:     }
34: }
```

```
UART#1
Hello World
Hello World
Hello World
Hello World
```

b) During interruption:

